

## Experiment 4

### Strategize Impedance Matching with a Short-Circuited Stub through Measurement of Location and Length of the Stub

#### Test Case 1

Optimize impedance matching using single stub matching by finding the location and length of the stub for load impedance  $Z_R$  varied from  $100 \angle 30^\circ \Omega$  to  $350 \angle 30^\circ \Omega$ , keeping the characteristic impedance  $Z_0 = 60 \Omega$  constant ( $f = 100 \text{ MHz}$ ).

#### Code -

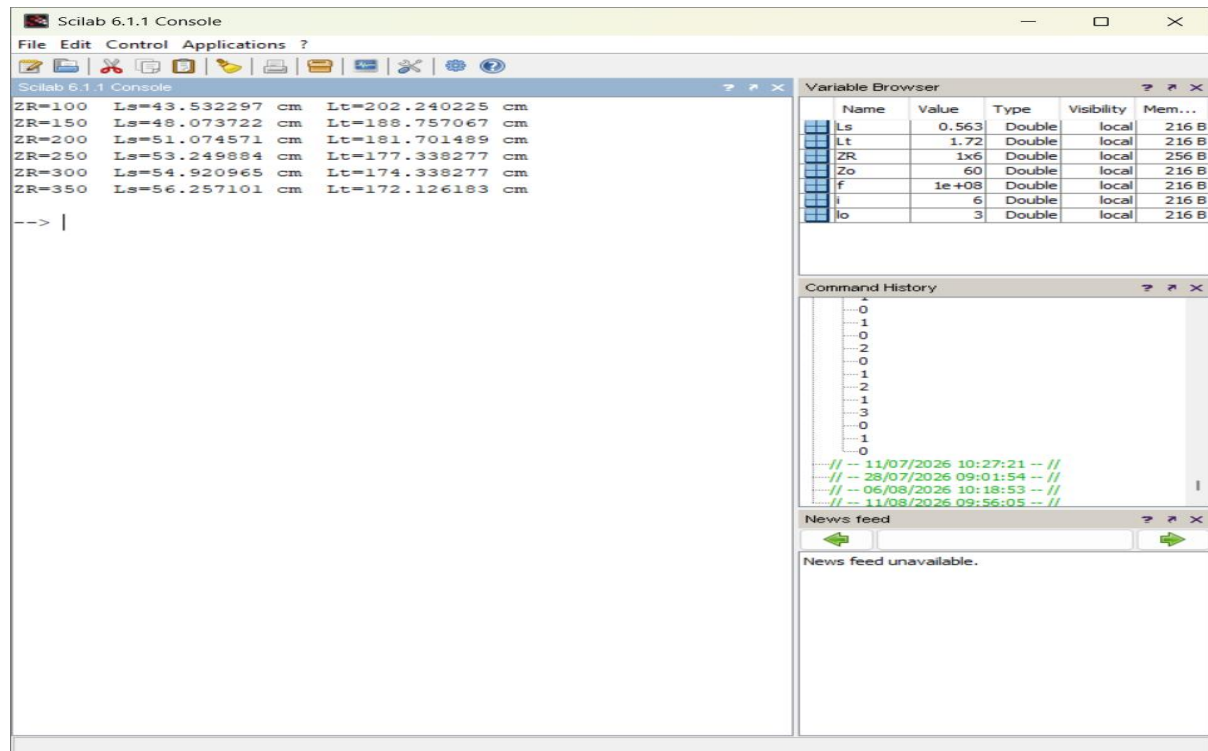
```
clear; clc;
Zo=60; f=100*10^6;
ZR=[100 150 200 250 300 350];
lo=300/(f*10^-6); // wavelength
for i=1:6
    Ls=(lo/(2*pi))*atan(sqrt(ZR(i)/Zo));
    Lt=(lo/(2*pi))*(%pi+atan(sqrt(ZR(i)*Zo)/(ZR(i)-Zo)));
    printf('ZR=%d Ls=%f cm Lt=%f cm\n',ZR(i),Ls*100,Lt*100);
end
```

#### Output:

$Z_0 = 60 \Omega$ ,  $f = 100 \text{ MHz}$ ,  $\lambda = 3 \text{ m} \rightarrow$  Variable:  $Z_R$

| $Z_R (\Omega)$ | Point of attachment<br>Ls (cm) - Sim | Stub length<br>(cm) - Sim | Ls (cm)<br>Theory | Lt (cm)<br>Theory |
|----------------|--------------------------------------|---------------------------|-------------------|-------------------|
| 100            | 43.53                                | 202.24                    | 43.53             | 202.24            |
| 150            | 48.07                                | 188.76                    | 48.07             | 188.76            |

|     |       |        |       |        |
|-----|-------|--------|-------|--------|
| 200 | 51.07 | 181.70 | 51.07 | 181.70 |
| 250 | 53.25 | 177.34 | 53.25 | 177.34 |
| 300 | 54.92 | 174.34 | 54.92 | 174.34 |
| 350 | 56.26 | 172.13 | 56.26 | 172.13 |



## Result:

Impedance matching using single stub matching was optimized by computing the stub location and length for varying load impedance ZR, keeping  $Z_o = 60 \Omega$  constant. Both the point of attachment and the stub length increase with ZR, though the rate of increase reduces at higher ZR.

## Test Case 2

Optimize impedance matching using single stub matching by finding the location and length of the stub for characteristic impedance  $Z_0$  ranging from  $600\ \Omega$  to  $1800\ \Omega$ , keeping the load impedance  $Z_R = 100\angle 30^\circ\ \Omega$  constant ( $f = 100\text{ MHz}$ ).

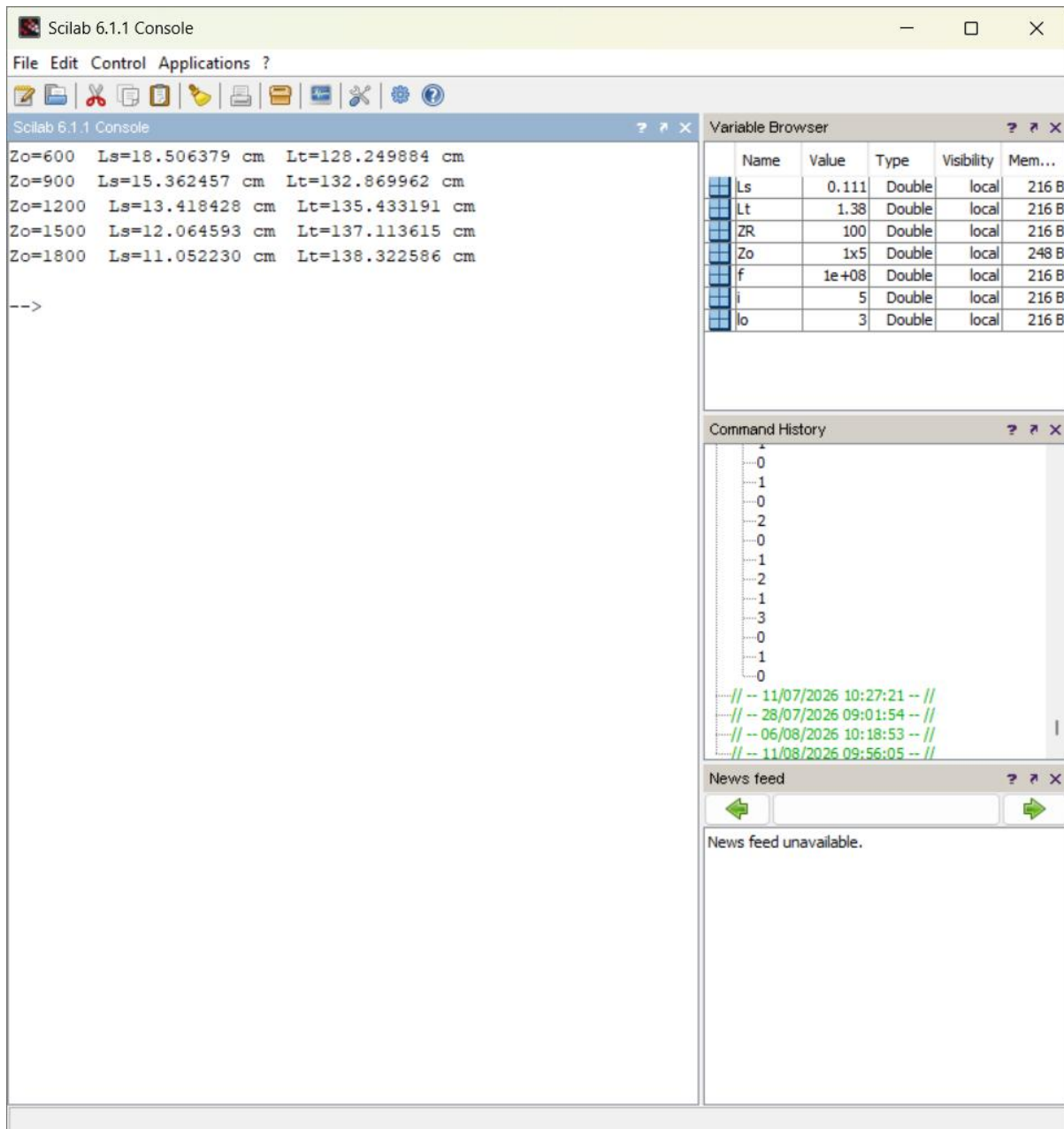
### Code -

```
clear; clc;
ZR=100; f=100*10^6;
Zo=[600 900 1200 1500 1800];
lo=300/(f*10^-6);
for i=1:5
    Ls=(lo/(2*pi))*atan(sqrt(ZR/Zo(i)));
    Lt=(lo/(2*pi))*(pi+atan(sqrt(ZR*Zo(i))/(ZR-Zo(i))));
    printf('Zo=%d Ls=%f cm Lt=%f cm\n',Zo(i),Ls*100,Lt*100);
end
```

### Output:

$Z_R = 100\ \Omega$ ,  $f = 100\text{ MHz}$ ,  $\lambda = 3\text{ m}$   $\rightarrow$  Variable:  $Z_0$

| $Z_0\ (\Omega)$ | Point of attachment<br>$L_s\ (\text{cm})$ - Sim | Stub length<br>$L_t\ (\text{cm})$ - Sim | $L_s\ (\text{cm})$ - Theory | $L_t\ (\text{cm})$ - Theory |
|-----------------|-------------------------------------------------|-----------------------------------------|-----------------------------|-----------------------------|
| 600             | 18.51                                           | 128.25                                  | 18.51                       | 128.25                      |
| 900             | 15.36                                           | 132.87                                  | 15.36                       | 132.87                      |
| 1200            | 13.42                                           | 135.43                                  | 13.42                       | 135.43                      |
| 1500            | 12.06                                           | 137.11                                  | 12.06                       | 137.11                      |
| 1800            | 11.05                                           | 138.32                                  | 11.05                       | 138.32                      |



## Result:

Impedance matching using single stub matching was optimized by computing the stub location and length for varying characteristic impedance  $Z_o$ , keeping  $Z_R = 100 \Omega$  constant. As  $Z_o$  increases, the required stub location moves closer to the load while the stub length increases.